

Code Localization as Sequential Set-Coverage: A Trainable Multi-Round Scaffolding for Small LLMs

Anonymous ACL submission

Abstract

Code localization refers to the task of identifying the files and functions that need to be edited to resolve a GitHub-style issue description. This is a critical step in automating software development and maintenance using coding agents. Strong localization is a prerequisite for strong code generation. However, localization requires inspecting large codebases, which consumes a large number of tokens, often exhausting the token budget. This is primarily because existing coding agents use terminal commands or tools designed to perform open-ended exploration of the codebase. Although this approach is feasible for large models with long contexts, it comes at the cost of huge token consumption and is almost impossible to fit in the context of smaller open source models.

To solve this, we propose a multi-agent system with a smarter codebase navigation scaffolding, that delegates token-heavy tasks of skimming through various code subgraphs to a smaller and cheaper post-trained agent (scorer) while working in unison with a larger and costlier agent to save its token consumption. We observe that such a collaborative system is not only cost effective but can augment any existing localization method and improve their performance across different benchmarks and programming languages.

1 Introduction

LLM coding agents increasingly resolve real GitHub issues by editing a repository directly, and benchmarks such as SWE-Bench now track how often they succeed (Jimenez et al., 2024b). A patch can only be correct, however, if it changes the right code. Before any edit, the agent must identify the files and functions an issue actually touches, the step known as *code localization*. This step is decisive and largely unrecoverable: once a wrong location enters the agent’s editable context, the repair stage reasons over the wrong code, and no later

reasoning recovers the correct patch (Jimenez et al., 2024b; Yang et al., 2024a). The strongest systems are built around this fact. Xia et al. (2024a) drop agentic interaction and resolve issues with a fixed localize-then-repair pipeline, crediting much of their accuracy to a hierarchical localization phase, and methods that improve the localization stage report corresponding gains in end-to-end resolution (Sohrabizadeh et al., 2025). Localization is therefore not a preprocessing convenience but the step that determines how well downstream repair can perform.

Yet localizing well is hard: a real repository holds thousands of files joined by tangled import, call, and inheritance edges, while an issue report describes symptoms in natural language that can sit far from the code that must change. Current systems close this gap in four ways. Reinforcement learning methods train the search policy itself: RepoSearcher pairs tool-integrated RL with rejection-sampled supervised fine-tuning (Ma et al., 2025), and SWE-RL scales RL over open software-evolution data (Wei et al., 2025). General-purpose agent scaffoldings give a capable LLM an open-ended action space with shell access, as in SWE-Agent and OpenHands (Yang et al., 2024a; Wang et al., 2025). Graph-exploration agents such as LocAgent let an LLM traverse a repository code graph, hopping freely between related code units (Chen et al., 2025b). Dense retrieval methods skip the agent altogether: SWERank scores code units with a bi-encoder retriever and a listwise LLM reranker (Reddy et al., 2025b).

The first three families pay for accuracy in tokens, and none of them bounds the cost. An RL agent, an open-ended scaffold, and a graph explorer all keep issuing tool calls and appending observations until they choose to stop, so a localization trajectory grows without a predictable ceiling. Under an OpenHands-style scaffolding, a single localization run can exceed 131K tokens (Section 4.4). A

model with a long context window and ample compute absorbs that cost; the open 7–14B code models most teams can realistically run do not. A Qwen-class coder with a 32K-token window overflows before the search terminates. Reinforcement learning compounds the problem: a multi-step episode returns only a sparse terminal reward, so adapting a small model to it takes many long and expensive rollouts. Dense retrieval is the one cheap option, but a single-step, order-centric ranker cannot revisit a discarded candidate or follow a structural edge, so it leaves the sequential, graph-structured side of the problem untouched. No existing method is at once bounded in token cost at inference and cheap enough to train on a small open model.

2 Related Work

Reinforcement learning for localization. RepoSearcher equips a repository agent with a curated tool set and trains it through rejection-sampled supervised fine-tuning followed by tool-integrated RL, reaching strong localization with a 32B model (Ma et al., 2025). SWE-RL scales RL over open software-evolution data and shows the learned reasoning transfers across software tasks (Wei et al., 2025). RepoNavigator narrows the agent to a single jump-to-definition tool and trains end-to-end with RL from a base model, so that a 7B agent can outperform substantially larger baselines (Zhang et al., 2025). In every case training optimizes multi-step episodes that return only a sparse terminal reward, so adaptation is costly in rollouts and compute. Our round decomposition exposes a dense per-round target that telescopes to recall@ L (Section 3.3), turning the same problem into ordinary supervised fine-tuning.

Agentic scaffolding. SWE-Agent and OpenHands wrap a strong LLM in a general-purpose environment with shell access and an open-ended action space, and they set much of the standard for end-to-end issue resolution (Yang et al., 2024a; Wang et al., 2025); Moatless provides a comparable tool-driven loop (Örwall, 2024). Agentless instead drops interaction and runs a fixed localize-then-repair pipeline, showing that a simple prompted procedure can rival full agents (Xia et al., 2024a). The interactive scaffoldings place no ceiling on trajectory length, so token cost is unbounded and a localization run easily overflows the context window of the open 7–14B models we target. Agentless is bounded, but its localization is a fixed prompted

pipeline, not a component a small model can learn to improve. Our scaffolding bounds the token cost of every round and stays trainable, so a small model can run the search within budget and be trained to localize better.

Graph-based exploration. A related line hands the LLM the repository’s structure. LocAgent builds a heterogeneous code graph and lets the model explore it through graph-traversal actions (Chen et al., 2025b), CoSIL has the model search a call graph it expands on the fly without a pre-built index (Jiang et al., 2025), and RepoGraph supplies a repository-level code graph for software-engineering agents (Ouyang et al., 2025). This work establishes that structural edges carry localization signal. The traversal, however, is unconstrained, so trajectory length and token cost again grow without a bound. We use the same structure through bounded neighborhood expansion around ranked seeds, so graph signal raises recall without inflating the budget.

Retrieval and reranking. Localization is also posed as retrieval. Lexical matching (Robertson et al., 1994) and learned code embeddings (Feng et al., 2020; Guo et al., 2021; Zhang et al., 2024a) score code units against the issue text, and SWERank sharpens this for issue localization with a bi-encoder retriever and a listwise LLM reranker (Reddy et al., 2025b); CoRet improves the retriever for code-editing queries (Fehr et al., 2025) and SpIDER incorporates repository structure into the dense representation (Chaudhari et al., 2025). These rankers are cheap and bounded, but the objective is per-query and order-centric, optimizing the order of one list rather than coverage of the gold set, and a single pass cannot revisit a discarded unit or follow a structural edge. SWERank+ adds a multi-turn agentic loop with a memory buffer (Reddy et al., 2025a), yet the loop is open-ended and the training signal stays a ranking objective. We recast localization as sequential set-coverage on a code graph and train an objective provably co-optimal with recall@ L (Section 3.4), keeping the bounded cost of retrieval while adding the set recovery and multi-round refinement that ranking omits.



Figure 1: The multi-round localization scaffolding. Each round consumes a bounded window of K ranked candidates, expands neighborhoods around C seed centers via bounded BFS on the code graph, and refines a fixed-length returned list L_r . Per-call token cost stays roughly constant across rounds.

3 Method

3.1 Problem Setup

3.2 Multi-Round Localization Scaffolding

The scaffolding processes a single issue q over R rounds. A retriever (dense, lexical, or their union) returns a ranked list of code units, and each round consumes the next K unseen entries of that list. At the start of round r , the scaffolding holds three pieces of state: the current returned list L_{r-1} (initialised to $\emptyset$ at $r=0$), the set of node identifiers $\mathcal{S}_{r-1}$ already shown to the model, and an offset rK into the ranked list. The loop terminates when the ranked list is exhausted or a maximum number of rounds $R_{\max}$ is reached. An optional confident-stop hook is left for the RL stage (Section 5); the supervised training of Section 3.4 does not predict it, so a trajectory always runs to $R_{\max}$ or exhaustion.

Each round combines a bounded window of newly-ranked candidates with a bounded graph-neighborhood expansion, then refines L ; Figure 1 sketches the loop.

Window. The scaffolding pulls the next K unseen entries of the ranked list starting at offset rK , skipping any node already in $\mathcal{S}_{r-1}$. We denote this window W_r . If W_r is empty the round terminates the loop.

Step A: seed-center selection. A single model call selects C seed centers from W_r , i.e. nodes whose graph neighborhoods are most likely to contain unseen gold units. If the model returns fewer than C valid identifiers we pad with the top-ranked nodes of W_r not yet returned, so Step A always emits exactly C centers.

Step B: neighborhood expansion. For each of the C centers, the scaffolding runs a depth- d BFS

along structural edges of the code graph (*contains* by default; ablations in Section 4.3 cover the four edge types). The neighbors are filtered to the next N unseen entries of the ranked list, so the candidate set the model sees stays bounded. A model call per center then selects the subset of these neighbors that must be modified, and the union across centers, $\mathcal{N}_r$, is deduplicated.

Step C: list refinement. A final model call sees the issue q , the current list L_{r-1} , the window W_r , and the deduplicated set $\mathcal{N}_r$, and selects L identifiers from $L_{r-1} \cup W_r \cup \mathcal{N}_r$ to form L_r . The scaffolding then records the newly-shown identifiers, advances the offset, and continues.

Three properties of this loop matter for what follows. *First*, every model call sees only W_r , a single neighborhood, or the deduplicated round pool, so the per-call token cost is roughly constant across rounds; a 32K context comfortably absorbs each call regardless of how many rounds the trajectory uses. *Second*, the loop offers a natural ablation: removing Steps A and B reduces it to a pure window-refine loop and isolates the contribution of graph exploration. *Third*, L_r at any round is a valid prediction, so the round structure exposes a per-round potential that we develop into a training target in the next section.

3.3 A Dense, Recall-Aligned Per-Round Target

3.4 Supervised Training Objective

4 Experiments

4.1 Setup

4.2 Main Results

4.3 Does the Coverage Term Help?

4.4 Efficiency and Scaling

5 Conclusion

Limitations

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A Proof of the Telescoping Identity

B Code Graph Construction

The scaffolding’s neighborhood expansion (Section 3.2, Step B) walks a per-repository code graph $\mathcal{G}=(V, E)$. We follow the four-edge schema of

Chen et al. (2025b) as implemented in Chaudhari et al. (2025), summarised here for completeness.

For each repository the graph is built from the source files written in the repository’s primary language; files in secondary languages are ignored. Nodes represent code entities at four granularities (directory, file, class, and function), and edges encode four structural relations: a *contains* edge linking each container to the code units it declares, an *invokes* edge between a function and each function it calls, an *imports* edge between a file and each module it imports, and an *inherits* edge between a class and each parent class it extends. Every non-directory node stores its source text alongside its identifier, so the model can see code when a node is shown.

We extract Python source files with the standard ast module and Java, JavaScript, and TypeScript files with Tree-sitter¹. The traversal of the repository’s directory tree adds directory nodes for navigation, skips files that contain no class or function definition, and emits one node per declared entity together with its containment, call, import, and inheritance edges. Multi-file class definitions and the more flexible declaration patterns of the non-Python languages are handled with language-specific Tree-sitter queries; we refer to Chaudhari et al. (2025) for the full list of cases. Graphs are constructed on demand at the start of a localization run and reused across the rounds.

For the experiments in Section 4, we use the graphs released with Chaudhari et al. (2025) for SWE-PolyBench and SWE-Bench Verified, and construct graphs from scratch for the SWE-Bench Lite dev split with the same pipeline.

C Scaffolding and Training Details

D Additional Results

E AI Disclosure

¹<https://github.com/tree-sitter>